

• Logical Operators

• AND statement: it allows us to combine multiple conditions

↳ All the conditions must be satisfied / met

↳ Commonly use the AND statement when building logical conditions for filtering data

Syntax:
 SELECT column 1
 FROM table_name
 WHERE condition 1 AND condition 2 AND condition 3

Students

student_id	name	course	age	city
1	Thabo	DA	22	JHB
2	Sarah	DS	19	PTA
3	John	DA	23	DBN
4	Lerato	DA	20	JHB
5	Musa	DS	25	JHB

DA = Data Analytics

DS = Data Science

Write an SQL query to give me the names of the students living in JHB that are doing data analysis

↓
 SELECT *
 FROM students
 WHERE city = 'JHB'
 AND course = 'DA'
 ↓

student_id	name	course	age	city
1	Thabo	DA	22	JHB
4	Lerato	DA	20	JHB

• OR statement

↳ combine multiple conditions

↳ one of the conditions must be met

Students

student_id	name	course	age	city
1	Thabo	DA	22	JHB
2	Sarah	DS	19	PTA
3	John	DA	23	DBN
4	Lerato	DA	20	JHB
5	Musa	DS	25	JHB

Write an SQL query to give me the names of the students living in JHB or the students doing DA

↓
 SELECT *
 FROM students
 WHERE course = 'DA' OR
 city = 'JHB'

Answer highlighted in pink

• IN Statement

↳ it is a way of filtering data by specifying the values of interest from a specific column

Syntax:
 SELECT *
 FROM table_name
 WHERE column IN ('value 1', 'value 2', 'value 3', ...)

No limit in the number of values you can specify

Students

student_id	name	course	age	city
1	Thabo	DA	22	JHB
2	Sarah	DS	19	PTA
3	John	DA	23	DBN
4	Lerato	DA	20	JHB
5	Musa	DS	25	JHB

Write an SQL query to give me people living in DBN & PTA

```
SELECT name, city
FROM students
WHERE city IN ('DBN', 'PTA')
```

ANSWER highlighted in pink

• NOT statement : to negate; to get the opposite result; the negative result

Students

student_id	name	course	age	city
1	Thabo	DA	22	JHB
2	Sarah	DS	19	PTA
3	John	DA	23	DBN
4	Lerato	DA	20	JHB
5	Musa	DS	25	JHB

Write an SQL query to give me people not living in DBN & PTA

```
SELECT name, city
FROM students
WHERE city NOT IN ('DBN', 'PTA')
```

• BETWEEN statement

↳ it allows us to filter the data based on a specific range of values

↳ it is an inclusive statement, this means that the lower & upper bounds of the BETWEEN statement are included

Syntax:
 SELECT column 1
 column 2
 column 3
 FROM table_name
 WHERE column BETWEEN lower bound AND upper bound

Students

student_id	name	course	age	city
1	Thabo	DA	22	JHB
2	Sarah	DS	19	PTA
3	John	DA	23	DBN
4	Lerato	DA	20	JHB
5	Musa	DS	25	JHB

Write an SQL query to give me the name and age of students between the ages of 20

and 23

```
SELECT age, name
FROM students
WHERE age BETWEEN 20 AND 23;
```